

Automated Counting of White-Tailed Deer in Vehicle-Mounted Thermal Video: Detector Benchmarking and a Staged Error Decomposition

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Abstract

Wildlife surveys often aim to estimate the abundance of animals in an area, yet automated pipelines are usually judged as object detectors, by how well they locate animals in individual frames rather than by how many distinct animals they count. We test whether detection accuracy predicts counting accuracy on thermal road-transect video. We introduce 32 fully annotated FLIR thermal road transects containing 236 individually tracked white-tailed deer, benchmark twelve detectors, and build TRACT, a counting pipeline whose error we decompose into detection, association and confirmation. The confirmation rule is fitted on training videos and frozen before evaluation on videos the detector never saw. Scored per animal rather than per box, detection is close to solved. The reported pipeline finds 89.2% of held-out animals, gives 79.5% a distinct track and counts 66.3%; the richest candidate pool we built reaches 98.8% and counts fewer still, at 62.7%. Every configuration that enriches the candidate pool makes the final count worse, and learned confirmation heads all lose to a three-parameter rule. In this survey, detection is not the bottleneck; confirmation is. At this corpus size it is limited by supervision rather than model capacity, because the extra candidates a better detector supplies are re-sightings of animals already counted.

Keywords: Thermal imaging, Wildlife survey, Animal counting, Object detection, Multi-object tracking, White-tailed deer

1 Introduction

Wildlife management rests on reliable estimates of abundance. White-tailed deer are among the most intensively managed ungulates in North America. Their density sets

hunting quotas, culling decisions and vehicle-collision mitigation, and the stakes extend beyond harvest. Surveillance for chronic wasting disease is most efficient where infection risk is highest [27]. Deer-to-human transmission of SARS-CoV-2 was identified in animals sampled during routine surveillance of exactly that kind [48]. Each of these decisions depends on knowing how many animals are present.

For deer, one established way of estimating abundance is to drive transects at night with spotlights or handheld thermal scopes. Road transects are practical but not neutral. Spotlight surveys along roads carry a known bias [64]. A road used as a transect line can also violate a central assumption of distance sampling, that animals are distributed uniformly with respect to the line, because deer may be drawn to or kept away from the road [16]. Thermal imaging extends the survey into darkness by detecting emitted heat, but it brings constraints of its own. Contrast falls as the ground approaches body temperature, vegetation hides animals, and a distant deer occupies only a few pixels [11]. Any automated survey inherits all of these at once.

Automating the survey is an obvious application of object detection, and the literature has largely treated it as one. From camera traps [45] to aerial imagery [30], systems are trained on annotated animal imagery and judged by per-image accuracy, typically mean average precision (mAP). The survey problem is regarded as addressed once that number is high. This paper argues, and then measures, that the two problems come apart. A detector is rewarded for finding animals in *frames*. A survey needs the number of distinct *animals*. Between the two sits a chain of decisions that no detection metric scores: associating detections into tracks, deciding which tracks are real, and deciding which are duplicates of an animal already counted.

That distinction fixes the yardstick we use throughout. A survey does not need a box placed precisely on an animal; it needs the animal noticed once. We therefore score the detection stage *per animal*, over the whole of an animal’s track, asking whether any frame produced a box touching it. We report this alongside the conventional per-box metrics rather than in place of them. Per-box mAP remains a valid measure of localization, but it answers a question a survey does not ask, and read as a survey metric it can place the difficulty in the wrong stage. Which stage limits a counting system is not something to assume. In sonar fish counting, substituting a perfect detector recovers almost all of the counting error, so there detection is the binding constraint [29]. A thermal deer transect differs in ways that could move the constraint elsewhere. The animals are faint, but each is visible over many consecutive frames as the vehicle passes, which gives association and confirmation far more evidence to work with. Settling the question needs an instrument that attributes error to each stage separately, applied to data annotated at the level of individual animals.

We study this on a corpus built for the purpose: 32 FLIR thermal road transects from four sites in Illinois, 521,930 frames at 640×512 and 60 fps, with every individual deer annotated as a distinct track. Because one annotated track is one animal, the corpus supplies detection, tracking and count ground truth at once, which is the property the decomposition depends on. Our central instrument is a three-stage decomposition of counting error. For each ground-truth animal we ask three questions. Is it REACHED, touched by at least one candidate track? Is it PRIMARY, covered by some

candidate better than that candidate covers any other animal, so that a one-count-per-track decision *could* count it? And is it finally COUNTED? The gaps between these three quantities attribute error to detection, to association and to confirmation, and Section 3.6 states the decomposition formally.

With the loss localized, we test both remedies a practitioner would reach for. We enrich the candidate pool by lowering the detector’s confidence threshold, removing the tracker’s track-initialization gate, relaxing non-maximum suppression (NMS), and adding an appearance term to association. We then replace the rule-based confirmation stage with learned alternatives of increasing capacity, under three evaluation protocols. In each case the question is the same: does the count improve?

The paper contributes three things. The first is the corpus: fully annotated thermal road transects with per-individual identity, which makes detection, tracking and counting evaluable against one set of annotations, released with the evaluation harness. The second is the decomposition, together with a held-out protocol that evaluates counting only on videos the detector never trained on. The third is a controlled test of where counting error arises and whether improving detection or confirmation reduces it. The remaining analyses support or qualify these three: the detector benchmark, the matching-criterion analysis, the sensitivity of the confirmation rule, the limits of appearance matching at 27 pixels, and the transfer between sites.

2 Related Work

2.1 Wildlife Survey from Imagery

Every field method for estimating abundance carries assumptions whose violation biases the result. Distance sampling fits a detection function to distances recorded along transects and extrapolates to density. The *Distance* software implements conventional, multiple-covariate and mark-recapture engines, each relaxing more assumptions, but even the most flexible still requires near-certain detection on the line, no responsive movement and error-free distances [61]. Buckland et al. [10] argue that camera, acoustic and spatial capture-recapture surveys are displacing classical distance sampling faster than statistical ecology is adapting. For unmarked populations, N-mixture models treat site-level population size as a latent Poisson variable estimated from replicated counts [55]. Spatial capture-recapture extends this to camera-trap encounters but requires individually identifiable animals [56], which thermal imagery at our resolution cannot supply. Iijima [25] reviews the estimators available for unmarked populations and concludes that none is universally applicable. Two recent corrections apply directly to road transects. Movement compensation corrects the detection inflation caused by animals encountering a moving observer [39]. Animals may also be attracted to or repelled by the road itself, which violates the uniform-distribution assumption [16].

Thermal cameras detect animals by emitted heat and so allow nocturnal survey where visual methods fail. Burke et al. [11] identify the binding constraints: contrast loss as ground temperature approaches body temperature, atmospheric absorption, vegetation occlusion, and roughly ten pixels on target for species identification. Their drone survey of riverine rabbits still needed ground confirmation of every sighting.

Karp [28] measured a 36% detection probability for hare leverets over 109 handheld searches, with vegetation height dominating. Ulhaq et al. [62] showed that transfer learning partly compensates for poor spatial resolution in airborne thermal video. Mowen et al. [40] classified animal poses from vehicle-mounted FLIR imagery at 82% accuracy while noting that a single-city dataset limits generalisation. Reviews of remotely piloted survey [24] and of monitoring methods for large mammals [49] reach the same conclusion: thermal sensors detect homeothermic animals well, but automated species identification from thermal imagery alone remains unreliable. McMahon et al. [37] compared UAS thermal survey against pellet-group counts for white-tailed deer and obtained overlapping but very wide intervals, 4–24 deer km^{-2} . They remain the closest published comparison for our species, and they report density rather than individual counts. Nazir and Kaleem [43] note that thermal work lags the RGB domain in dataset scale and standardisation.

Aerial and underwater counting studies show the same reliance on per-frame evaluation. Brown et al. [8] tested detection across ground sample distances from 0.2 to 1.0 m px^{-1} and found that image detail, not network capacity, sets the limit. Infantes et al. [26] built an end-to-end pinniped survey from flight planning to automatic body-size measurement. Kline et al. [32] tracked zebra and giraffe from an autonomous uncrewed aerial vehicle (UAV) but analysed herd centroids rather than individuals, and Mpouziotas et al. [41] reached 91.3% AP on drone bird footage evaluated per frame only. Kilfoil et al. [31] found that both human and CNN counts of sea cucumbers underestimate at high density. Tarling et al. [60] applied density regression with uncertainty quantification to sonar imagery of fish schools, using self-supervised pre-training to compensate for a small labelled set. Automated bird detection has also been evaluated for airport strike management [47].

Deep learning for camera traps was catalysed by Norouzzadeh et al. [45], who matched crowd-sourced labels on 3.2 million Snapshot Serengeti images. Beery et al. [3] then showed that models trained at one camera location fail at unseen ones, and Kellenberger et al. [30] found the same failure across terrain in aerial survey. Singh et al. [59] documented limited transfer from natural to man-made environments, and Beery et al. [4] showed that synthetic renders reduce error for rare classes only. Liu et al. [35] released a twelve-year longitudinal video dataset and confirmed that web-image pretraining transfers only partially to camera-trap footage. Adam et al. [1] designed a similarity-aware split for a re-identification benchmark to prevent the near-duplicate leakage that inflates reported accuracy, the same contamination our held-out protocol (Section 3.7) excludes. Throughout this literature the unit of analysis is the trigger event, and individual identity is assumed unknown unless the species carries distinctive markings.

One absence runs through all of it. These methods treat imperfect detection as a parameter to be estimated, not as a sequence of pipeline stages whose losses can be separated. The abundance estimators model it as a detection function, the imaging studies evaluate per frame, and the camera-trap literature takes the trigger event as its unit. We found no thermal or camera-based wildlife study that annotates individual animals as distinct tracks across continuous video, and none that reports a distinct-individual count evaluated against per-individual ground truth. That is the

gap this paper addresses. Its ground truth is itself subject to imperfect detection, since an annotated animal is one a reviewer could see in the footage. The decomposition therefore attributes the losses of an automated pipeline relative to what is visible in the video, not to the true number of animals present (Section 5.3).

2.2 Detection and Tracking at Small Scale

COCO defines small objects as area below 32^2 [34], and wildlife at range falls well under it. Wang et al. [67] show that IoU is extremely sensitive to localisation error at small scale. A one-pixel diagonal shift drops the IoU of a 6×6 box to 0.53 and a four-pixel shift to 0.06, whereas a 36×36 box keeps 0.90 and 0.65 under the same shifts. Their Normalized Wasserstein Distance models boxes as Gaussians and gains 6.7 AP on tiny-object benchmarks. Zhu et al. [77] added a transformer prediction head and attention to target the drone scale distribution. Section 3.5 makes the same sensitivity exact for our box size, and Section 4.2 shows its consequence: at 27 pixels the strict threshold discards detections a counting task would accept, and $\text{IoU} \geq 0.50$ re-ranks the detectors relative to a counting criterion.

Both one- and two-stage families appear in our benchmark: Faster R-CNN [52] as the two-stage baseline, ATSS [72] and TOOD [19] as anchor-assignment refinements, RTMDet [36] as a modern convolutional one-stage design, and the YOLO line [51, 65, 66] as the practical default. Query-based detectors struggle on very small objects: DETR [13] also converges slowly, DINO [71] repairs the convergence, and RT-DETR [76] brings the family to real-time inference. Recent reviews trace the YOLO line from v1 to v12 [57] and compare v8 through v11 architecturally, noting that some blocks are unchanged across versions and that missing publications for several releases complicate reproducibility [22]. Our animals are a median 29×24 pixels, squarely where these differences matter. Section 4.1 reports how each family transfers under identical preprocessing and splits.

Tracking-by-detection dominates. SORT [5] and DeepSORT [69] established the Kalman-plus-association template, ByteTrack [74] showed that associating low-confidence detections recovers substantial recall, and BoT-SORT [2] added camera-motion compensation and an appearance term. Our camera moves at $25\text{--}90 \text{ km h}^{-1}$, so global motion compensation is not optional. We use BoT-SORT with sparse optical-flow compensation and ablate its appearance term in Section 4.4. Evaluation has moved toward higher-order tracking accuracy (HOTA), which separates detection from association accuracy, and Valmadre et al. [63] argue that global metrics mask local failures. MOTRv2 [75] shows that even end-to-end transformer trackers benefit from strong detection priors. Our REACHED/PRIMARY/COUNTED decomposition parallels the detection/association split but adds a third axis, confirmation, that these metrics do not measure. Multi-object tracking benchmarks score identity *preservation*, not whether an object should be counted at all.

2.3 Counting, Identity and Evaluation

Density regression, pioneered by MCNN [73], estimates per-image counts without detection, but the estimate is per frame and does not deduplicate across frames. That

suits dense scenes with hundreds of overlapping instances. Ours is the opposite regime: sparse instances, each of which must be resolved as an individual because survey statistics are computed per animal. Video-level counting is structurally different, since it requires detection, association, and a rule that decides whether an object is *new*. Dolokov et al. [17] exploit a known maximum animal count to reduce identity switches in laboratory mice, but report tracking quality rather than a unique-individual count. Our contribution is to show that in the sparse regime the detection-then-count framing is not wrong so much as *mis-attributed*. The detector is near-saturated, and the accuracy a practitioner would try to buy with a better backbone is lost downstream.

Individual identification from appearance is established for species with high-contrast markings. Photo-identification networks reach high accuracy on coat patterns [38], deformable transformers reach 85% mAP on cross-camera tiger data [33], and WildlifeReID-10k spans 33 species and 10,000 identities [1]. All operate on high-resolution crops of distinctively patterned animals. Section 4.11 tests whether this transfers and finds that it does not: at a median 27-pixel animal, appearance carries less identity signal than a two-number size descriptor, and an ImageNet backbone scores *below* plain geometry. We therefore make no individual-identification claim and do not use appearance matching in the reported system.

Nguyen et al. [44] show that detection rankings fluctuate with the IoU threshold and propose formal requirements that average precision does not satisfy. LRP Error [46] separates localisation from classification quality and yields a deployment threshold. Cai and Vasconcelos [12] document the high-quality detection paradox, where raising the training IoU threshold causes overfitting as positives vanish, and Boundary IoU [14] shows that mask IoU over-penalises small objects. Our finding that $\text{IoU} \geq 0.50$ re-ranks seven of eleven detectors by two or more places, and that the best AP_{50} model counts fewer animals than the one we adopt, instantiates exactly the concern Nguyen et al. raise. On the calibration side, Rahman et al. [50] predict per-frame mAP from internal features to monitor deployment without ground truth. Tarling et al. [60] show that uncertainty quantification improves both training and downstream biological decisions, which is the role our calibrated per-track posterior plays in Section 3.4.

The closest methodological comparison is the Caltech Fish Counting benchmark [29], which poses the same problem in sonar: count distinct individuals in continuous video, not detections per frame. Its conclusion is the opposite of ours. Substituting a perfect detector recovers almost all of the counting error, so in that setting detection is the binding constraint. Sonar imagery has a low signal-to-noise ratio, and the reported failures are generalization failures of detection across deployments. A thermal deer at 27 pixels is faint but consistently present, and a vehicle transect gives every animal tens to hundreds of frames of visibility with strong ego-motion cues for association. What generalizes from this paper is therefore not “detection is solved” but the *instrument*: a staged decomposition that reports which stage binds, so that a practitioner in a new modality can measure rather than assume. Section 4.9 reinforces the point, since detection recovers some ground as a bottleneck once sites change.

That framing also identifies the literature our confirmation stage should read against. Duplicate suppression at the detection level is mature: Soft-NMS re-scores

rather than discards overlapping boxes [6], and Confluence replaces the IoU criterion with a proximity measure that behaves better on small objects [58]. All of it operates *within* a frame. The duplicates that limit counting here are between *tracks*, separated in time, on the same animal, and Section 4.8 tests whether the frame-level tools reach that far. Calibration is the other relevant thread. Modern neural networks are systematically overconfident [20], which is why Section 3.4 reports a calibrated per-track posterior rather than a raw score, and why Section 4.9 finds that an absolute score threshold does not survive a change of site.

2.4 Positioning

Table 1 Where this work sits. The closest comparables in thermal wildlife survey, automated aerial counting, and animal re-identification. “n/r” means the study does not report the quantity.

Work	Modality	Platform	Taxa	Per-ind. ID	Video	Count?	Split
Burke 2019 [11]	Thermal	UAV	Rabbit	No	No	No	n/r
Karp 2020 [28]	Thermal	Handheld	Hare	No	No	No	n/r
Ulhaq 2021 [62]	Thermal	Aircraft	Multiple	No	Yes	No	n/r
McMahon 2021 [37]	Thermal	UAV	W. deer	No	No	Density	n/r
Corcoran 2019 [15]	Thermal	UAV	Koala	No	No	No	n/r
Mowen 2022 [40]	Thermal	Vehicle	Multiple	No	Yes	No	n/r
Brown 2022 [8]	RGB	UAV	Livestock	No	No	No	Yes
Norouzzadeh 2018 [45]	RGB	Camera trap	48 spp.	No	No	No	Site
Beery 2018 [3]	RGB	Camera trap	Multiple	No	No	No	Location
Kellenberger 2018 [30]	RGB	Aircraft	Multiple	No	No	No	Region
Liu 2023 [35]	RGB	Camera trap	Endangered	No	Yes	No	Domain
Adam 2025 [1]	RGB	Mixed	33 spp.	Yes	No	No	Similarity
Tarling 2022 [60]	Sonar	Fixed	Fish	No	Yes	Density	n/r
Dolokov 2023 [17]	RGB	Overhead	Mice	Yes	Yes	No	n/r
This work	Thermal	Vehicle	W. deer	Yes	Yes	Yes	Video

Table 1 makes the gap specific. Its “per-individual ID” column asks whether each animal is annotated as its own track rather than as anonymous per-frame boxes. The last two columns isolate the gap: whether a count of distinct animals is reported end to end, and whether training and evaluation are separated by video or site rather than by frame. Thermal datasets with per-individual track identity over continuous video are essentially absent. Existing thermal work annotates per-frame detections or static images, and the RGB video datasets that carry individual identity are either photo-identification collections of still crops [1] or laboratory settings with a known number of animals [17]. Train–test separation by video or site is frequently unclear in the thermal and aerial literature. Our own measurement quantifies what that costs: evaluating on all 32 videos rather than the 13 held out inflates coverage by 15.5 points, from 66.3% to 81.8% (Table 4). Most distinctively, no prior work isolates *which stage* of a detect–track–confirm pipeline loses animals. The literature treats detection quality as the binding constraint; our decomposition shows that it is not here, and that each richer candidate pool we built made the final count worse. The closest conceptual parallel

is the detection/association split within HOTA, which measures identity preservation but not whether an object should be counted at all.

3 Materials and Methods

3.1 Dataset: Corpus, Annotation and Splits

The corpus is 32 nocturnal road-transect videos recorded with a vehicle-mounted FLIR thermal camera at four sites in southern Illinois, eight videos per site. Frames are 640×512 single-channel thermal at 60 fps. The vehicle travels at 25–90 km/h, so the background translates continuously and the same animal is visible for a few seconds at most. Figure 1 shows representative frames with their annotation and Table 2 summarizes the corpus. The top row is what a detector receives: in the third and fourth panels the animals are a few dozen pixels of slightly warmer texture against thermal clutter. The median animal is 29×24 px, roughly the smallest box in the third panel. Thermal transects carry low and drifting global contrast, so every frame is normalized with contrast-limited adaptive histogram equalisation (CLAHE) [53] before use, identically at training and inference and for every model reported here. This single preprocessing choice is worth more than any architectural one we tested: test mAP50 falls from 0.519 to 0.299 without it. The figures in this paper therefore show normalized rather than raw frames, because they are the pixels the detectors consume.

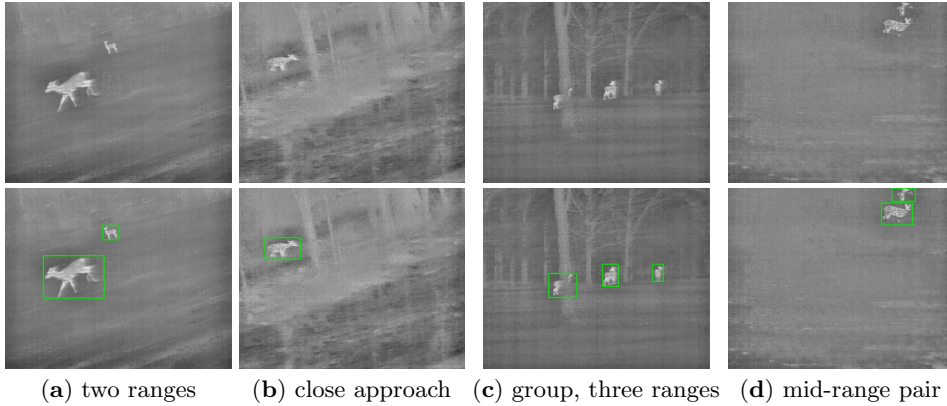


Fig. 1 Dataset examples. Top row: raw thermal frames after the CLAHE normalization the detector is trained and evaluated with. Bottom row: the same frames with their ground-truth boxes. Panels show the *large* end of the corpus for legibility at print scale.

Two frame counts appear in this paper and they measure different things. The 32 videos contain 521,930 frames, and *every* deer in *every* frame is annotated in the Computer Vision Annotation Tool (CVAT) as a member of an individual track. That annotation is the 236 tracks and 21,646 boxes quoted above, and it is what “fully annotated” means here. At 60 fps consecutive frames are near-duplicates, so the detector dataset of Table 2 is an extraction from the annotation: 18,914 frames sampled

Table 2 Dataset composition. 32 thermal road-transect videos from four Illinois sites, 640×512 at 60 fps, a single annotated class. Splits are by video and site-stratified. Rows count the frames *extracted* for detector training, not the 521,930 frames of the corpus; the complete track annotation is 21,646 boxes, of which the 12,903 here fall inside the extracted frames.

Split	Videos	Frames	With Deer	Empty	Boxes
Train	19	12,852	5013	7839	7547
Val	4	2337	935	1402	2564
Test	9	3725	1490	2235	2792
Total	32	18,914	7438	11,476	12,903

around annotated activity together with explicit negatives, carrying the 12,903 boxes that fall inside them. The 21,646 figure is the size of the ground truth and the 12,903 the size of the detector’s training signal. The counting evaluation of Section 4.3 always runs against the former, on whole videos decoded end to end. Of the 18,914 extracted frames, 7438 contain at least one deer and 11,476 are explicit negatives; background frames matter, since a survey system must not fire on warm rocks or structures.

The 236 animals are unevenly distributed across the four sites: SHB contributes 132 (56% of the corpus), TON 51, SHW 38 and MAS 15. One TON track is marked outside the frame on every frame, the animal being occluded throughout, so it can never be detected. **235 animals are therefore detectable, and every detection and counting denominator in this paper is 235**, while 236 remains the count of annotated individuals. The imbalance is not incidental. SHB is the dense-group site where most of the counting error concentrates (Section 4.3), and MAS is small enough that its per-site figures should be read as indicative only. Every individual deer is annotated as a distinct *track*, and this is the property the paper depends on. One track is one animal, so the same annotation serves as per-frame detection ground truth, as association ground truth and as count ground truth, the number of tracks in a video. In Section 4.11 it also serves as identity labels.

We split by *video*, never by frame, and stratify by site so that all four sites appear in each split. The deployment target is these same sites, and a frame-level split would leak near-duplicate frames across the boundary. The result is 19 training videos (12,852 images), 4 validation (2337) and 9 test (3725), allocated by animal count to roughly 65/19/16%. Throughout the paper, **“held out” means the 13 validation and test videos the detector never trained on**, containing 83 of the 236 deer. Section 3.7 shows why the distinction matters: counting accuracy measured over all 32 videos, 19 of which trained the detector, is 15.5 points higher than on held-out video alone.

3.2 Object Scale

The animals are small and the distribution is one-sided. The median box is 29×24 pixels ($\sqrt{\text{area}} = 26.8$), 71% of boxes fall below the COCO [34] small-object threshold of 32^2 , and *zero* boxes reach the COCO-large category. Resolution is therefore the strongest single accuracy lever, which motivates the ablation in Section 4.1. The one-sidedness has a second consequence, developed in Section 4.10: no box in the training

split exceeds 96 pixels, and detector performance falls on the few animals that appear larger. The upper tail of the scale distribution is as under-served as the lower one, and it is far less often checked.

3.3 Counting Pipeline

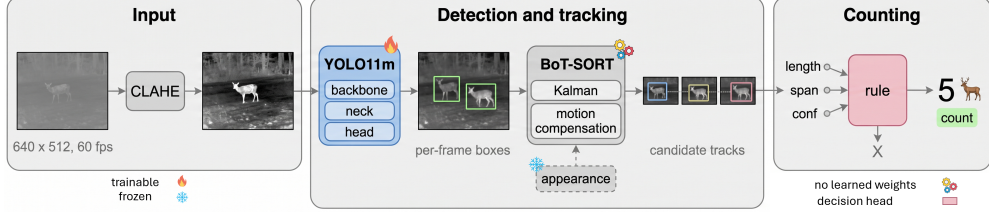


Fig. 2 The TRACT counting pipeline. Thermal frames are contrast-normalized, detected per frame, associated into candidate tracks, and filtered by a three-parameter confirmation rule; the count is the number of accepted tracks. Dashed components were ablated and are not used. Frames are illustrative.

Figure 2 shows the system, which we call **TRACT** (Thermal Road-transect Animal Counting and Tracking). Every component is standard; the assembly of contrast normalisation, detection, motion-compensated tracking, orphan recovery and a calibrated confirmation rule is what is arranged for counting rather than for detection. Detections from YOLO11m at 640px are associated by BoT-SORT [2] with sparse-optical-flow global motion compensation, which is mandatory here because the whole frame translates as the vehicle moves. The result is a set of candidate tracks. One point of terminology belongs here, because the association stage is easily mistaken for something it is not. **The reported system runs with BoT-SORT’s appearance term disabled and does not attempt to recognise individual animals.** It associates detections into tracks within a single video using motion and geometry, which prevents an animal from being counted twice in one transect. It could not recognise individuals in any case, since Section 4.11 shows that at a median 29×24 pixels a two-number size descriptor out-performs a pretrained appearance backbone. We call this *track association* throughout. Where appearance appears in this paper it is an *ablation* (pool G of Section 4.4) or a probe of how much identity signal the imagery carries (Table 12), never a component of the reported counts.

Only the detector is gradient-trained. The tracker’s Kalman filter, motion compensation and association carry no weights, and the confirmation rule’s three parameters are fitted by grid search. The pipeline also has no stochastic component at inference. Run three times on the same GPU over four held-out videos holding 55 of the 83 held-out animals, it produced byte-identical detections, tracks and counts. A count is therefore a fixed function of the video, and repeating a run cannot reduce its error. Across platforms, one benchmarked detector evaluated on an NVIDIA GH200, with a different processor architecture and different PyTorch and CUDA builds, reproduced its A100 keyframe AP_{50} of 0.670; counts themselves were compared only on identical

hardware. The detector is not the one with the best AP_{50} , and the choice is deliberate. Section 4.2 selects on the precision-weighted $F_{0.5}$ instead. The confirmation stage must reject tens of thousands of false candidates against roughly two hundred true ones, so it pays far more for a false positive than for a false negative. That section also verifies the choice end to end: the highest- AP_{50} model run through this same pipeline counts fewer animals.

The confirmation stage decides which candidate tracks are real, distinct animals, and the count is the number it accepts. Let a candidate track τ consist of $m(\tau)$ detections spanning $s(\tau)$ seconds, with per-frame detector confidences ordered as $c_{(1)} \geq c_{(2)} \geq \dots \geq c_{(m)}$. Its track score is the mean of its k highest confidences,

$$\bar{c}_k(\tau) = \frac{1}{k} \sum_{j=1}^k c_{(j)}, \quad k = 5, \quad (1)$$

and the rule accepts τ when three conditions hold at once,

$$a(\tau) = \mathbf{1}[m(\tau) \geq m^*] \mathbf{1}[s(\tau) \geq s^*] \mathbf{1}[\bar{c}_5(\tau) \geq c^*], \quad \hat{c}_v = \sum_{\tau \in \mathcal{T}_v} a(\tau), \quad (2)$$

where $\mathbf{1}[\cdot]$ is the indicator function, $\mathcal{T}_v$ is the candidate set of video v and $\hat{c}_v$ is its count. The score is a property of the *track*: a low-confidence frame is never discarded, it continues to extend the track, and a candidate is rejected only when its five best frames average below c^* . The three thresholds are not set by hand. They minimise counting error over the 19 detector-training videos $\mathcal{V}_{\text{fit}}$,

$$(m^*, s^*, c^*) = \arg \min_{(m, s, c) \in \mathcal{G}} \frac{1}{|\mathcal{V}_{\text{fit}}|} \sum_{v \in \mathcal{V}_{\text{fit}}} |\hat{c}_v(m, s, c) - N_v|, \quad (3)$$

where N_v is the annotated number of animals in v and the grid $\mathcal{G}$ is $m \in \{1, 2, 3, 5, 8, 12, 20\}$ frames, $s \in \{0, 0.1, 0.25, 0.5, 1, 2\}$ s and $c \in \{0, 0.15, 0.25, \dots, 0.85\}$, 378 cells in all. The values are frozen before any held-out video is scored (Section 3.7). The search selects $m^* = 20$, $s^* = 0$ and $c^* = 0.65$. At that operating point the span condition imposes nothing, since $s \geq 0$ accepts a track of any duration, so the rule is in effect two parameters. We keep s because it is the parameter the search uses to compensate when c is lowered (Section 4.6), and Fig. 8 shows the whole search. We additionally recover *orphan* detections, those the tracker returns without a track identity, by linking detections within a temporal gap and a scale-normalized distance into pseudo-tracks. Without this, single-frame animals are discarded entirely; with it, 17 animals are recovered.

3.4 Calibrated Per-Animal Confidence

We calibrate a per-track posterior for “this track is on a real animal” and report it with every counted animal. **92.1% of counted animals carry confidence ≥ 0.80** (mean 0.965, area under the ROC curve 0.893), and 793 low-confidence tracks are flagged

for human review rather than auto-counted. The calibrator is **isotonic regression fitted on the validation videos only** [70]: a monotone step function $p = \phi(\bar{c}_5)$ that maps the confirmer’s raw score to a probability by least squares under a monotonicity constraint. No held-out video contributes to the fit. We prefer isotonic to Platt scaling because the raw score is not sigmoidal in the log-odds, and a monotone step function makes no shape assumption. The cost is that the output takes a finite number of levels, 46 of them on held-out video. We also propagate the per-track probabilities into an uncertainty on each video’s count. If the accepted tracks of video v carry probabilities $p_1, \dots, p_{\hat{c}_v}$ and are treated as independent, the number of real animals among them is Poisson-binomial [23], with

$$\text{Var}[\hat{c}_v] = \sum_{i=1}^{\hat{c}_v} p_i (1 - p_i). \quad (4)$$

That quantity is narrower than it looks. It expresses uncertainty about whether the tracks already accepted are real animals and says nothing about animals that were never detected or never accepted. On held-out video its nominal 95% band therefore contains the true count in only one of thirteen videos. It should not be read as an interval on abundance, which would need a model of the probability that an animal is detected and counted at all.

Table 3 reports whether the calibrated number behaves like a probability on videos the calibrator never saw. A track scores as positive when it is the *primary* track of a real animal; duplicates count as negatives, which is the decision the counting stage makes. We use the two standard summaries [20]. With the n scored tracks partitioned into $B = 10$ equal-count bins $\mathcal{B}_b$ by predicted probability, the expected calibration error [42] and the Brier score [7] are

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{n} |\bar{y}_b - \bar{p}_b|, \quad \text{BS} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2, \quad (5)$$

where $y_i \in \{0, 1\}$ is the label, p_i the calibrated probability, and $\bar{y}_b$ and $\bar{p}_b$ the mean label and mean probability in bin b . Over all 2211 held-out candidate tracks the ECE is 0.004 and the Brier score 0.017, against 0.029 for a constant predictor at the base rate. That aggregate is generous, because 74% of candidates sit at the isotonic floor where no decision is ever taken. Restricted to the 316 tracks in the decision region the ECE is 0.034 and the Brier score 0.093 against 0.148. The second is the one a practitioner should read. One caveat belongs with them: isotonic regression saturates, so the top level reports 1.000 on 13 held-out tracks whose true positive rate is 0.846. Confidence at the very top of the scale is optimistic by roughly 15 points, and the flag-for-review band, not the top of the scale, is where the calibrated number earns its keep.

Raw detector confidence cannot serve this purpose. No track in the corpus exceeds 0.90 and the median counted track sits at 0.74, because the animals are 27 pixels, so the requirement is only meaningful against a calibrated posterior. Figure 3 shows why a score must be per track rather than per frame. The raw confidence varies by 0.15 across frames of the same animal (Animal B, 0.66–0.81) even where every frame

Table 3 Reliability of the per-track posterior on the 13 held-out videos, against the isotonic calibrator fitted on validation videos only. Positives are primary tracks; duplicates are negatives. The constant-predictor column is the Brier score of always predicting the base rate, so the gap is the skill the posterior adds.

Scope	Tracks	Base rate	ECE	Brier	Constant
All held-out candidates	2211	0.030	0.004	0.017	0.029
Decision region ($p > 0.01$)	316	0.180	0.034	0.093	0.148
Fit videos, for reference	4797	0.029	0.003	0.017	0.029

is a clean detection, so no single frame’s value characterises the track. Every counted animal is also exported as a contact sheet of its own track, one panel per sampled frame with elapsed time and per-frame score, so a human can verify any count against the frames that produced it.

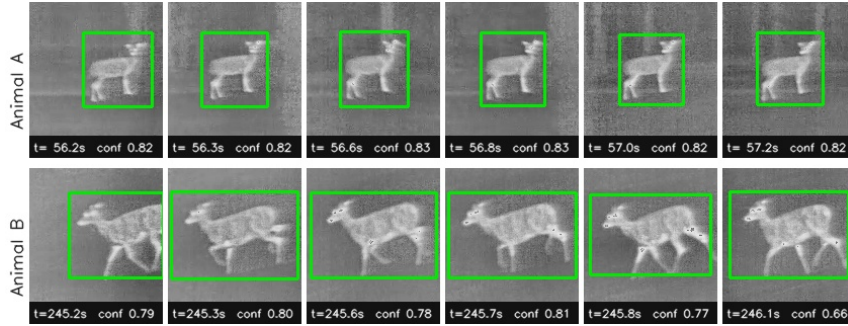


Fig. 3 Per-animal evidence, two individuals. Each row is one animal’s own track, one panel per sampled frame, with elapsed time and raw per-frame detector confidence beneath. Both animals are from videos the detector never trained on.

3.5 Matching Criterion

A detection metric asks whether a predicted box *localizes* an animal; a survey asks whether the animal was *noticed*. Intersection over union (IoU), the standard localization score [18, 34], diverges sharply from the survey question on a 27-pixel target, and the divergence can be made exact.

Proposition 1 (IoU under a centre offset) *Let a predicted box and a ground-truth box be axis-aligned squares of equal side s whose centres differ by $d < s$ pixels along one axis. Then*

$$\text{IoU}(d) = \frac{s-d}{s+d}, \quad \text{so that} \quad \text{IoU}(d) \geq t \iff d \leq s \frac{1-t}{1+t}. \quad (6)$$

Proof The intersection is a rectangle of sides s and $s - d$, with area $s(s - d)$. The union is the sum of the two areas minus the intersection, $2s^2 - s(s - d) = s(s + d)$. The ratio gives the first equality, and solving $(s - d)/(s + d) \geq t$ for d gives the second. $\square$

At the corpus median $s = 27$, Eq. (6) says that a centre error of more than 9 pixels fails $\text{IoU} \geq 0.50$ and one of more than 3.9 pixels fails $\text{IoU} \geq 0.75$. A box offset by six pixels is therefore scored as both a false positive and a false negative at $\text{IoU} 0.5$, while for counting it has correctly found the animal. We therefore evaluate under four matching rules: $\text{IoU} \geq 0.50$, $\text{IoU} \geq 0.30$, centre-in-box and *any overlap*. The last is the counting criterion, reported as presence or counting recall and never as mAP, with the strict criterion alongside it everywhere. The criterion is validated against centre-in-box, an independent counting-oriented rule that accepts a prediction only when its centre falls inside the animal and so rejects oversized boxes outright. The two agree to within 0.032 F1 across all eleven models, so the conclusions rest on two criteria agreeing rather than on one being chosen. Precision is reported alongside recall throughout, which further constrains box quality.

3.6 Decomposing Counting Error

Let G_v be the ground-truth animals in video v , $N_v = |G_v|$, and $\mathcal{T}_v$ the candidate tracks the pipeline produces. For each animal $g \in G_v$ we define three indicators:

- $R(g) = 1$ (REACHED) if g is touched, on at least one frame, by at least one candidate. If an animal is not reached, no downstream stage can recover it; this measures **detection and tracking**.
- $P(g) = 1$ (PRIMARY) if, among the candidates touching g , the one that covers g on the most frames is not also the best cover for some other animal. Two animals sharing a single best-covering candidate are indistinguishable to any rule that emits one count per track; this measures **association**.
- $K(g) = 1$ (COUNTED) if the confirmation stage accepted a candidate covering g . This measures **confirmation**, and it is the end-to-end result.

The stage rates are the means of these indicators over the animals under evaluation, $\rho = \overline{R}$, $\pi = \overline{P}$ and $\kappa = \overline{K}$. An accepted candidate is a candidate, so $K \leq R$ and $P \leq R$ by construction, and in every configuration we report $\kappa \leq \pi \leq \rho$, so the three read as a funnel whose steps attribute error to the three stages. We report all three for every configuration; reporting only the last, as is conventional, makes a detection miss and a confirmation miss look identical.

The video-level error has an exact decomposition of its own. Credit each accepted track to at most one animal, the one it covers on the most frames, and each animal to at most one accepted track, and let A_v be the number of animals credited in video v . A track that covers two animals credits only one of them, so the pair contributes one credited animal and one miss. Write $D_v = \hat{c}_v - A_v \geq 0$ for the accepted tracks

that credit no new animal (duplicates of an animal already credited, and tracks on no animal) and $M_v = N_v - A_v \geq 0$ for the animals with no credited track.

Proposition 2 (Signed error is duplicates minus misses) *For every video, $\hat{c}_v - N_v = D_v - M_v$, and $\min(\hat{c}_v, N_v) = A_v + \min(D_v, M_v)$.*

Proof Both follow from $\hat{c}_v = A_v + D_v$ and $N_v = A_v + M_v$. Subtracting gives the first identity; taking the minimum of the two and factoring out A_v gives the second. $\square$

Proposition 2 is the bookkeeping the rest of the paper relies on. A count is unbiased only when the duplicates it admits equal the animals it misses, which is the trade Section 4.6 measures, and every intervention that recovers animals is judged by whether it lowers M_v faster than it raises D_v . The aggregate metrics over a set of videos $\mathcal{V}$ are

$$\text{MAE} = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} |\hat{c}_v - N_v|, \quad \text{Bias} = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} (\hat{c}_v - N_v), \quad \kappa_{\text{cap}} = \frac{\sum_v \min(\hat{c}_v, N_v)}{\sum_v N_v}, \quad (7)$$

where κ_{cap} is the *counted* fraction, capped per video so that over-counting in one video cannot mask misses in another. By Proposition 2 it exceeds the identity-matched fraction $\sum_v A_v / \sum_v N_v$ by exactly $\sum_v \min(D_v, M_v) / \sum_v N_v$. The capped aggregate thus credits a video for predicting the right *number* even when some accepted tracks do not land on the animals, and Section 4.3 shows the two differ by 10 points. Counting quantities carry 95% intervals throughout. Proportions over the 83 held-out animals use the Wilson score interval [68], and means over the 13 held-out videos use a t -interval on the per-video values. These are sampling intervals over the corpus, and with 83 animals they are wide: a three-animal difference is not resolvable. We deliberately do *not* attach intervals to the detection metrics of Section 4.1. Each architecture was trained once, so a bootstrap over the test images would give a narrow interval that describes test-set sampling while saying nothing about the training variance that separates one architecture from another.

3.7 Held-Out Protocol

Counting runs process all 32 videos, but the detector trained on 19 of them, and the confirmation rule is conventionally swept on the same videos it is reported on. Both leaks inflate the result. Our protocol closes both: **sweep the rule on the 19 detector-training videos, freeze it, and report on the 13 held-out videos**. Learned confirmers are trained and thresholded under the identical split, so the comparison in Section 4.7 is like-for-like.

The protocol is worth 15.5 points (Table 4): counting accuracy measured over all 32 videos is 81.8%, and on the 13 held-out videos it is 66.3%. Four of the 13 held-out videos are the detector’s *validation* split. No gradient was computed on them, but they were used for checkpoint selection, so they are held out in the sense that matters for the confirmation rule and only partly held out for the detector. They carry 45 of the 83

Table 4 Why the held-out protocol matters. Evaluating on all 32 videos, as an in-sample pipeline evaluation would, reports 81.8% coverage. On video the detector never trained on it is 66.3%. Both figures use the capped per-video fraction of Section 3.6; the uncapped ratio of predicted to true animals reads 89.0% and 69.9% respectively, and is the more flattering of the two because over-counting one video offsets misses in another.

Evaluation Scope	Videos	Deer	Coverage
Detector-training videos	19	153	90.2%
Held out (val + test)	13	83	66.3%
All 32 (the default)	32	236	81.8%

animals. We state this rather than reporting on the 9 test videos alone, for two reasons. The confirmation rule, the subject of the paper, saw none of the 13, and splitting the held-out set further would leave 38 animals to carry every counting conclusion.

3.8 Box Size as a Range Proxy

The corpus records no range to the animals, so box size stands in for it in Sections 4.10 and 5.3. The relation follows from the pinhole camera [21]: an animal of physical length L at range r subtends

$$s = \frac{fL}{r} \quad (8)$$

pixels, where f is the focal length in pixels. Box size is therefore inversely proportional to range for animals of similar size, and it conflates range with body size and posture, a caveat that applies wherever the proxy is used. The proxy also yields a null expectation for how many animals should be annotated at each size, which Section 5.3 uses to test the ground truth itself for range-dependent censoring.

Proposition 3 (Availability law for box size) *Suppose animals of length L are distributed with uniform density λ per unit ground area inside the camera’s horizontal field of view of angle θ , and every animal in view is annotated. Then the expected number of annotated animals with box size in $[a, b)$ is*

$$E[a, b) = \frac{\lambda\theta(fL)^2}{2} \left(\frac{1}{a^2} - \frac{1}{b^2} \right), \quad (9)$$

so the density of animals per unit box size falls as s^{-3} . If instead the animals lie in a strip of fixed width w along the road, $E[a, b) = \lambda w f L (1/a - 1/b)$, a density falling as s^{-2} .

Proof The ground in view between ranges r and $r + dr$ is a circular sector of area $\theta r dr$, holding $\lambda\theta r dr$ animals in expectation. By Eq. (8), $r = fL/s$ and $|dr| = fL s^{-2} ds$, so the expected number with box size in $[s, s + ds)$ is $\lambda\theta(fL)^2 s^{-3} ds$. Integrating from a to b gives Eq. (9). In a strip of width w the ground between r and $r + dr$ has area $w dr$, and the same substitution gives $\lambda w f L s^{-2} ds$, whose integral is $\lambda w f L (1/a - 1/b)$. $\square$

The constant in front is unknown, but it cancels in ratios. Fitting it to the number of animals observed in a reference size range predicts the number expected in every

other range, which is how Section 5.3 applies the result. Distance sampling makes the same uniform-density assumption for animals relative to a transect line [9, 61], and a road transect can violate it [16], which is why the strip law is carried along as the more conservative alternative.

4 Results

4.1 Detection Benchmark

Twelve architectures spanning two-stage, one-stage and query-based designs were trained on the identical site-stratified split with identical preprocessing. That bounds how much of the counting error any choice of backbone could account for. Eleven appear in Table 5. Deformable DETR exhausted GPU memory at epoch 5 and was not rerun at a smaller batch, so every count of models in this section is eleven.

Table 5 Detection on the held-out test split (3725 images, 2792 boxes) at 640 px and confidence 0.25, ordered by F1; best per column in bold. The column groups are three different questions, not three views of one. The strict criteria ask whether a box *localizes* an animal, any overlap asks whether it *touches* one, and the last column asks whether the *animal* was found at all in any frame of its track. Only the last is the quantity a survey depends on, and it is the one on which the eleven models are indistinguishable.

Model	AP ₅₀	F1, strict criteria			Any overlap				Per animal
		IoU _{≥.50}	IoU _{≥.30}	Centre	P	R	F1	F _{0.5}	Animals found
YOLOv8m	0.459	0.502	0.603	0.628	0.780	0.572	0.660	0.727	0.906
ATSS R50	0.380	0.441	0.556	0.648	0.683	0.635	0.658	0.673	0.962
TOOD R50	0.396	0.442	0.572	0.623	0.683	0.621	0.650	0.670	0.979
YOLO11m	0.460	0.535	0.623	0.638	0.897	0.503	0.645	0.775	0.949
RT-DETR-L	0.434	0.497	0.595	0.610	0.643	0.605	0.623	0.635	0.983
YOLO12m	0.508	0.506	0.583	0.595	0.929	0.452	0.608	0.767	0.864
YOLOv9m	0.506	0.527	0.581	0.592	0.917	0.450	0.604	0.759	0.821
YOLOv10m	0.510	0.527	0.575	0.583	0.874	0.453	0.597	0.737	0.885
DINO R50	0.365	0.425	0.520	0.548	0.560	0.572	0.566	0.562	0.983
Faster R-CNN R50	0.418	0.376	0.475	0.497	0.393	0.723	0.509	0.433	0.970
RTMDet-m	0.466	0.339	0.411	0.437	0.325	0.779	0.459	0.368	0.996

F1 rises monotonically across Table 5 as the matching rule relaxes, uniformly rather than favouring particular architectures (Section 3.5). Little separates the models. AP₅₀ spans 0.365–0.510, with the top three within 0.004, across designs as different as Faster R-CNN, DINO and YOLO12m. That narrow spread is the signature of a data-limited rather than an architecture-limited regime. Figure 4 shows it is not an artefact of under-training: within the budget each run either plateaus or passes its peak, and the plateaus sit on top of one another. Two patterns in that figure matter later. Precision settles near 0.65–0.70 while recall settles near 0.40–0.45 in every panel, the precision-heavy operating point Section 4.2 argues the counting stage needs. Several models peak early and then decline, so we report the best checkpoint rather than the last.

The absolute values are low for two reasons, neither of them a model failure. The first is Proposition 1: at the corpus median of 27 pixels a four-pixel centre error already fails IoU 0.75, and an error of more than nine pixels fails 0.50. The second is that these metrics count boxes rather than animals. An animal visible for 100 frames contributes 100 ground-truth boxes, and one found in half of them scores 0.5 recall even though counting requires a single detection. The 11,476 background frames contribute nothing either way, since precision, recall and F1 have no true-negative term.

The last column of Table 5 closes that gap. Measured per animal rather than per box, recall runs from 0.821 to 0.996. *Every* architecture we trained finds between 82% and 99.6% of the 235 detectable individuals in at least one frame, including models whose box recall sits near 0.45, and the full pipeline reaches 82 of 83 on held-out video (Section 4.3). Over 235 animals these are the paper’s tighter estimates; YOLO11m’s 98.7% carries a 95% interval of ± 2.4 points. At the level a survey cares about, detection is very nearly solved *on this corpus*, and the animals that go uncounted are lost after it. Section 4.9 tests how far that survives a change of site and finds that it weakens. The column also re-ranks the table a third time. RTMDet-m leads at 0.996 and every mmdetection model places above every YOLO variant. That is not an argument for adopting one, because their coverage is bought with false positives that the confirmation stage cannot absorb (Section 4.4).

A third pattern belongs to individual rows. YOLO12m’s 0.929 precision against 0.452 recall looks self-contradictory, and it is not. The two numbers describe one conservative operating point: of YOLO12m’s errors at confidence 0.25, 1530 are missed boxes and 96 are false ones, so **94% of its error is silence rather than mistake**. The distinction matters downstream, because the two failures cost the confirmation stage very different amounts.

Table 6 The precision–recall split is a threshold choice, not a capability limit. The two detectors with the highest $F_{0.5}$ in Table 5, under the presence criterion, at the conventional reporting threshold of 0.25 and at 0.10, the threshold at which the counting pipeline runs its detector. Rows are ordered by the best F1 either threshold achieves.

Model	Conf.	Precision	Recall	F1	TP	FP	FN
YOLO12m	0.25	0.929	0.452	0.608	1262	96	1530
	0.10	0.711	0.653	0.681	1824	741	968
YOLO11m	0.25	0.897	0.503	0.644	1404	162	1388
	0.10	0.777	0.576	0.661	1608	462	1184

Table 6 gives the sensitivity. At 0.10 the same YOLO12m weights score 0.711 precision and 0.653 recall, so the low recall in Table 5 is a property of the reporting threshold rather than of the model. Nor is 0.25 the operating point of the system this paper describes. **The counting pipeline runs its detector at confidence 0.10**, where the recall of YOLO11m rises from 0.503 to 0.576 and that of YOLO12m by 20 points; 0.25 is reported only because it is the conventional threshold for a detection table. The matching rule understates recall further. Relaxing $\text{IoU} \geq 0.50$ to presence

cuts YOLO12m’s false positives from 307 to 96, while its misses fall only from 1741 to 1530. A looser criterion can credit a box placed a few pixels off an animal, but it cannot recover an animal the detector never fired on. The trade is also asymmetric, since a detector already at high recall has nowhere left to go. Dropping RT-DETR-L to 0.10 buys 231 additional true boxes and costs 7433 false ones, reducing precision to 0.187, a candidate pool no confirmation stage can work with.

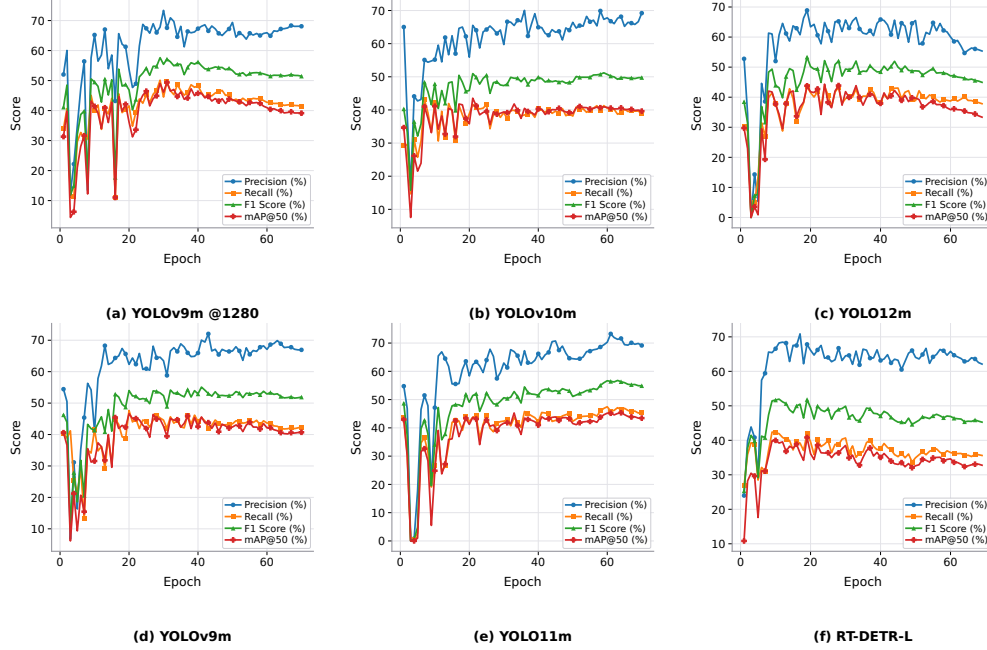


Fig. 4 Validation curves for the six strongest detectors, ordered by test AP_{50} and truncated at epoch 70 so the curves share a range. Drawn from the training log each run wrote; no curve is smoothed or re-fitted. The y -axes are not shared. The five mmdetection models are absent because their validation loop logs only the COCO evaluator’s output, which carries no scalar precision or recall.

The query-based detectors underperform on every criterion, although for DINO R50 this is a failure to converge rather than a settled comparison. Trained for 36 epochs, three times its native schedule, its validation score was still oscillating when training ended. Its 0.365 AP_{50} comes from the best validation checkpoint, below plain Faster R-CNN. RT-DETR-L, the real-time member of the family, recovers only to 0.434.

4.2 Choosing a Detector for Counting

The narrow spread in aggregate scores conceals a substantial difference in operating point. YOLO12m and YOLOv9m sit at 0.929 and 0.917 precision against roughly 0.45 recall. RTMDet-m and Faster R-CNN R50 occupy the opposite end, reaching 0.779 and 0.723 recall at 0.33 and 0.39 precision. Since a matching criterion determines

how errors of each type are counted, rank depends on which criterion is applied. Between the two orderings in Table 5, seven of eleven models move at least two places. YOLOv10m falls from 2nd to 8th and ATSS R50 climbs from 8th to 2nd, a six-place exchange between a precision-tuned and a recall-tuned model. Selecting a detector on its detection metric does not select the model that counts best.

Both AP_{50} and F1 weight precision and recall equally, which does not reflect the cost structure of the stage that consumes these detections. The confirmation step must reject tens of thousands of false candidates against roughly two hundred true ones (Section 4.3), so a false positive costs it far more than a false negative. Section 4.4 shows that enlarging the candidate pool lowers the final count. We therefore summarize with the precision-weighted F_β score [54] at its conventional value $\beta = 0.5$,

$$F_{0.5} = \frac{(1 + 0.5^2) PR}{0.5^2 P + R} = \frac{1.25 PR}{0.25 P + R}, \quad (10)$$

in the last column of Table 5. YOLO11m ranks first at 0.775, ahead of YOLO12m at 0.767 and YOLOv9m at 0.759. That margin of 0.008 makes the top three effectively equivalent; the substantive separation is from the recall-oriented models at 0.368–0.433. We adopt **YOLO11m at 640 px** as the production detector, and every counting result in this paper uses it.

Resolution is the one lever that behaves unpredictably, and it does not favour a larger input. Moving YOLOv10m from 640 to 1280 px lowers it on every metric, from 0.543 to 0.525 F1 and 0.510 to 0.469 AP_{50} , with AP-small falling from 0.117 to 0.093. YOLOv9m moves the other way over the same change, from 0.547 to 0.595 F1 and 0.506 to 0.516 AP_{50} . At 640 the two architectures are within 0.004 AP_{50} ; at 1280 they diverge by 0.047. Extra resolution therefore buys precision and recall on some architectures and costs them on others, and on neither does it buy the smallest animals. These figures are the detector validator’s own metrics at $IoU \geq 0.50$, so they are not comparable with the any-overlap columns of Table 5. The gain does not survive the pipeline. YOLOv9m at 1280 px is the strongest detector this section produced, so we ran it through the complete counting pipeline under the protocol of Section 3.7. It counts *fewer* animals: MAE 2.54 and 54 of 83 held-out animals, against 2.38 and 55 of 83 for YOLO11m at 640 px. An advantage in detection accuracy becomes a small deficit in counting accuracy. It is the first direct evidence for the divergence the rest of the paper characterizes.

4.3 End-to-End Counting

Table 7 is the end-to-end result: **MAE 2.38 ± 1.29 animals per video and 66.3 $\pm$ 10.7% of held-out animals counted**, with a bias of -1.92 ± 1.57 (95% intervals; Section 3.6). The system under-counts systematically and over-counts almost never. For management purposes that is the safe direction, since a survey that never inflates its estimate is preferable to one that occasionally does.

Table 8 gives the breakdown per video, and reading a row left to right attributes that video’s loss to a stage. The error concentrates in dense-group footage. SHB contributes 9 of the 25 net missing animals from three videos. Its group-heavy NShelbyRd

Table 7 End-to-end counting. The held-out row is the honest number: the rule is swept on the 19 detector-training videos, frozen, and applied to the 13 held-out ones. Bias is consistently negative: the system under-counts and essentially never over-counts, which is the conservative direction for a wildlife survey.

Scope	MAE	Bias	Counted (%)
Fit videos (detector trained on these)	1.53	−0.05	90.2
Held out (13 videos)	2.38 ± 1.29	-1.92 ± 1.57	66.3 ± 10.7
All 32 videos (optimistic)	1.88	−0.81	81.8

Table 8 Per-video breakdown on the 13 held-out videos. GT is the annotated number of individual animals; Reach and Prim. are the decomposition of Section 3.6; Count is the number the pipeline reports and Err its signed error. Reading a row left to right attributes that video’s loss to a stage.

Video	Site	GT	Reach	Prim.	Count	Err
NShelbyRd (blue)	SHB	27	22	16	20	−7
GolfDr	SHB	12	12	10	14	+2
NWolfCreek (orange)	SHB	9	9	9	5	−4
GiantCityRd	TON	8	7	7	3	−5
Robinson	SHW	7	7	7	7	0
Melvin	SHW	5	5	5	1	−4
AquaCultureRd	TON	3	1	1	0	−3
N25thBlue	MAS	3	3	3	3	0
NMarseilles	MAS	3	3	3	1	−2
OikosRd	TON	2	2	2	1	−1
TouchofNature	TON	2	1	1	0	−2
ChipsRd	TON	1	1	1	1	0
SIron	SHW	1	1	1	2	+1
Total		83	74	66	58	−25

transect alone accounts for 7, shedding 5 animals to detection (27→22) and a further 6 to association (22→16), the largest single loss in the corpus. Only two videos over-count, GolfDr and SIron, and both do so with complete reach and primary counts, so their error is confirmation accepting spurious tracks. Two videos return a count of zero. AquaCultureRd holds 3 animals and reaches 1; TouchofNature holds 2 and reaches 1. Both are *detection* failures, with animals at the sensor’s limit near the 20px median of the never-detected group in Section 4.10, so no adjustment to the counting rule recovers them. They also show why a per-video error of 3 and 2 animals understates how these rows read. On a transect holding three animals, missing all three leaves the count at zero, and the mean absolute error of 2.38 conceals that a small-count transect can be off by 100% while contributing little to the mean.

Under the identity-matched criterion of Section 3.6, animals credited with their own *accepted* track, the figure is $47/83 = 56.6\%$. By Proposition 2 the 10-point gap is $\sum_v \min(D_v, M_v)$, eight animals of coincidental agreement in which accepted tracks that land on no animal offset animals that were missed. The capped aggregate is our

headline figure, for comparability with the counting literature; the identity-matched figure is reported alongside it because the two answer different questions.

4.4 Improving Candidate Generation Makes Counting Worse

We enriched the candidate pool in three configurations and measured the effect on the decomposition and on the final count. Pool E lowers the detector threshold and removes the tracker’s initialization gate, and pools F and G each add one further change to E, so the three are nested rather than independent. Figure 5 shows the decomposition for each pool, Fig. 6 orders the same pools by how many candidates they generate, and Table 9 gives one row per intervention, with the pools labelled C, E, F and G throughout. Each intervention does what it was designed to do:

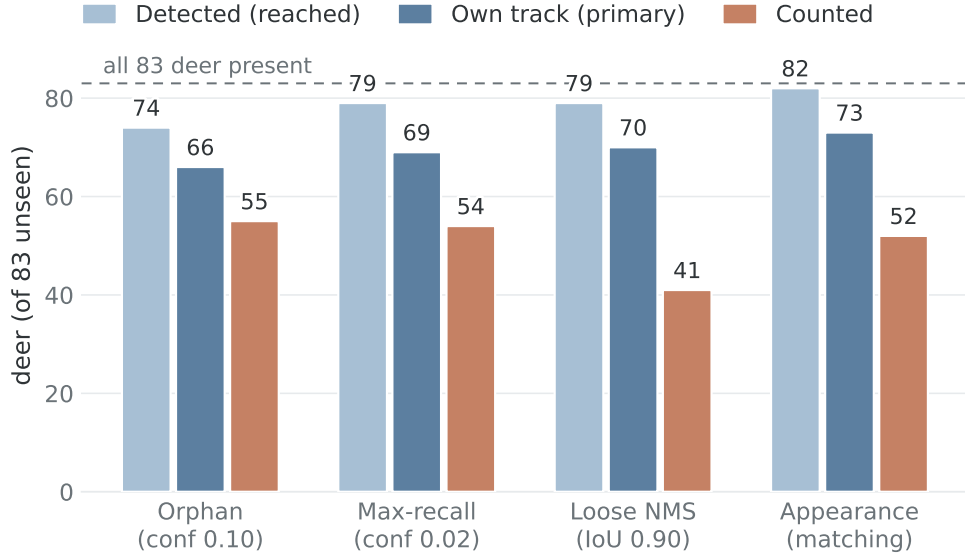


Fig. 5 Where held-out animals are lost, by candidate pool. Of the 83 deer in the 13 videos the detector never trained on, up to 82 are found by *some* candidate track and 73 acquire a distinct track of their own, but only 52–55 are counted. Every bar is one of four candidate-generation configurations; improving the pool moves the blue and green bars up and the orange bar *down*.

- **Lowering detector confidence** 0.10→0.02 and **removing the tracker’s track-initialization gate** raised REACHED from 74 to 79. Lowering confidence *alone* recovered zero additional animals. The gate was the binding constraint: BoT-SORT will extend a track from a low-confidence detection but will not *start* one, so faint animals were invisible to counting regardless of detector threshold.

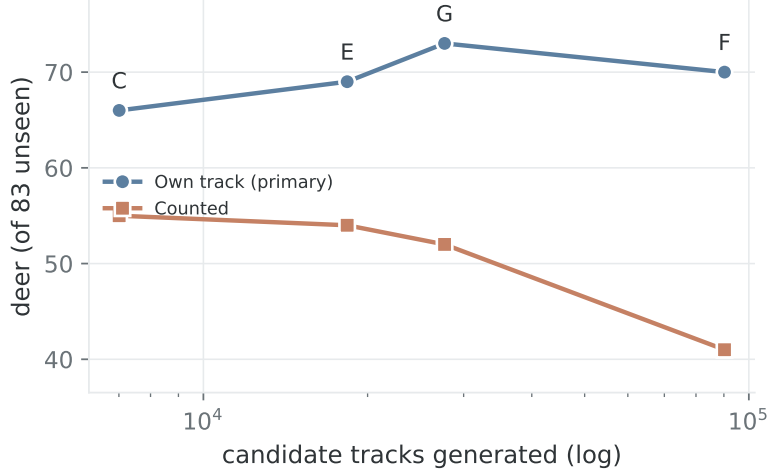


Fig. 6 The inversion. As candidate generation improves, more animals become reachable and fewer are counted. Both series are animals out of 83 and share one axis.

- **Loosening NMS** to IoU 0.90 addressed animals standing close enough that one box suppressed another; it added 4 PRIMARY at the cost of $5\times$ the candidates.
- **Appearance matching** produced the best pool by every measure: 82 of 83 animals reached and 73 with their own track, at 27,679 candidates, fewer than a third of pool F’s for four more PRIMARY. It merges fragments of one animal rather than merely adding candidates.

And the count falls, monotonically in candidate volume: 55, 54, 52, 41. With 83 animals a difference of two or three is not resolvable. The 95% interval on pool C’s 55 is ± 9 animals, and pools E and G lie inside it. No single pairwise comparison is significant, and we do not claim one. The evidence is the *direction*: every richer pool counted fewer animals than the baseline, and pool F’s 41 of 83 falls outside pool C’s interval. Because the pools are nested, that consistency is not a formal test, and we attach no probability to it. The mechanism is the ratio the confirmation stage sees. Each intervention adds far more negatives than positives, pool G has 219 positive tracks against 27,460 negatives, and a rule tuned to survive that ratio must be aggressive enough to reject real animals along with the noise.

4.5 Pipeline Ablations

Table 9 changes one stage per row, and the pattern is uniform: *no change to detection, tracking or association improves the count*. Appearance matching produces the best candidate pool by a wide margin: 82 of 83 animals reached and 73 given their own track, at less than a third of the loose-NMS candidate volume. It still counts three

Table 9 Pipeline ablation on the 13 held-out videos. Each row changes one stage of the full pipeline in the first row, all others held fixed. Pools C, E, F and G are the four candidate-generation configurations referred to by letter throughout; “—” marks a row that is not one of them. The appearance-matching row is a tested variant, not the reported configuration; the first row is. The last two columns mark, against the full pipeline, whether the intervention enriched the candidate pool and whether it improved the count: ✓ better, ✗ worse, – unchanged. Every row that enriches the pool still loses count.

Pool	Configuration	Candidates	reached	primary	counted	MAE	Richer pool	Better count
C	Full pipeline (YOLO11m@640, orphans, 3-param rule)	7008	74/83	66/83	55/83	2.38	–	–
—	detector → YOLOv9m @1280 px (best AP ₅₀)	1496	67/83	62/83	54/83	2.54	✗	✗
—	– orphan recovery	711	66/83	59/83	52/83	2.46	✗	✗
E	+ track-init gate removed, conf 0.02	18,349	79/83	69/83	54/83	2.54	✓	✗
F	+ NMS IoU 0.50 → 0.90	90,239	79/83	70/83	41/83	3.46	✓	✗
G	+ appearance matching	27,679	82/83	73/83	52/83	2.69	✓	✗
—	rule → temporal transformer	7008	74/83	66/83	49/83	2.85	–	✗

fewer animals than the baseline. The only row that changes the confirmation stage is the only place a substantial loss appears without a pool change, and Table 11 expands that comparison across four methods and two pools. One entry from the wider set is worth isolating. Lowering detector confidence from 0.10 to 0.05 added 3807 candidates and recovered *zero* additional animals, which ruled out the detector threshold and pointed us at the tracker’s initialization gate.

Figure 7 shows what the association stage does on held-out video under the two conditions that make counting fail. The first is overlap. When two animals stand close enough that their boxes intersect, a tracker that leans on spatial proximity will hand both to one identity, and the pair is counted once. The first two panels and the centre panel of the lower row each contain such a pair, and in every case the animals are carried as separate tracks. The first two panels are the same scene 0.3 s apart, with five animals tracked and five counted in both; tracks 3035 and 3041 keep their identities across the pair because motion compensation removes the camera’s contribution before association. The second condition is low contrast. In the centre panel of the lower row two animals of 42 and 50 px sit against an almost uniform thermal field, overlapping and close to indistinguishable. A human reader would hesitate to call them animals at all. They are still carried as two tracks. The lower-left panel shows the range spread a single frame can contain, 87 px and 22 px animals at once, with the calibrated score falling as the animal recedes.

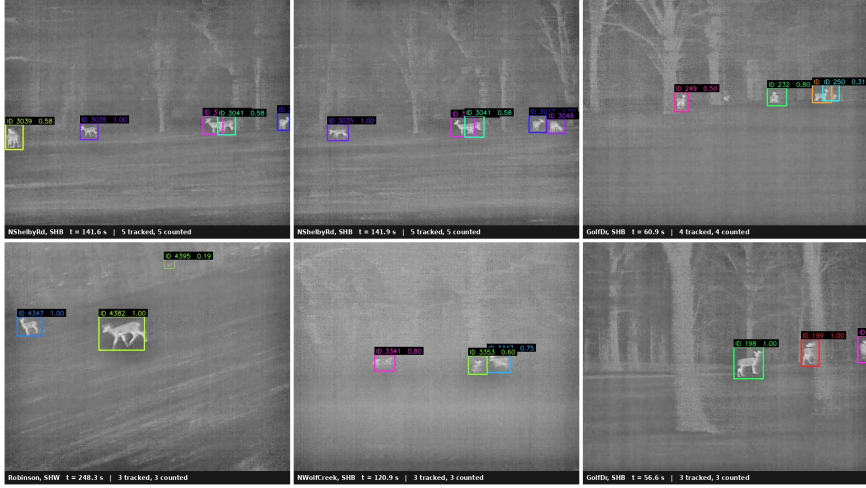


Fig. 7 Tracking through overlap and low contrast. Six frames from four held-out transects. Box colour encodes track identity and is stable over time; each label gives the track ID and its calibrated per-track score. The first two panels are the same scene 0.3s apart; the other four are separate scenes.

4.6 What the Confidence Condition Buys

The confirmation rule’s parameters come from the grid search of Eq. (3), and the confidence condition $c \geq 0.65$ is the one a reader is most likely to challenge. The detector did find those animals, and discarding a candidate because its five best frames average below 0.65 looks like throwing away recall that was already paid for. Measured directly, the condition is not a recall filter at all.

Table 10 Deleting the confidence condition, all else fixed. The published operating point ($m \geq 20$, $s \geq 0$) is held and only c moves; every column is the 13 held-out videos, 83 animals. Lowering c recovers real animals, raising coverage 23 points, and admits duplicates faster, so the system passes through approximate unbiasedness at $c \approx 0.45$ and ends over-counting by 28 animals at $c = 0$. The published $c = 0.65$ minimizes MAE; it does not maximize coverage, and Section 5 argues that for a survey this is the right trade-off.

c	Predicted	Counted	MAE	Bias	Over	Under
0.00	111	89.2%	3.54	+2.15	37	9
0.15	111	89.2%	3.54	+2.15	37	9
0.25	108	85.5%	3.77	+1.92	37	12
0.35	99	80.7%	3.69	+1.23	32	16
0.45	81	72.3%	3.38	−0.15	21	23
0.55	70	71.1%	2.69	−1.00	11	24
0.65	58	66.3%	2.38	−1.92	3	28
0.75	24	28.9%	4.54	−4.54	0	59
0.85	0	0.0%	6.38	−6.38	0	83

Table 10 holds the other two parameters at their published values and moves only c . Three things follow. First, **the objection is half right**. Removing the condition raises held-out coverage from 66.3% to 89.2%, the largest coverage gain from any single change in this paper; none of the four candidate-generation interventions of Section 4.4 raised coverage at all. Second, **they do not come back alone**. The pipeline predicts 111 animals where 83 exist. Over-counts rise from 3 to 37 while under-counts fall only from 28 to 9, so MAE rises from 2.38 to 3.54 and the bias flips from -1.92 to $+2.15$. In the terms of Proposition 2, 53 additional tracks are accepted, of which 19 credit animals that were previously missed and 34 are duplicates or false tracks, so each recovered animal costs roughly two duplicates. Third, **the condition is a duplicate suppressor rather than a detector threshold**. A fragment sitting on an animal that has already been counted is a true detection, so nothing in the question “is this a deer?” separates it from a first sighting. Sustained high confidence across a track is the cheapest signal that does, because a duplicate fragment is typically short and low-scoring where the animal’s primary track is neither. That is why c cannot be relaxed to recover recall: recall and duplicate suppression are bought with the same parameter. Section 5.3 takes up the bias column: the MAE-optimal setting is deliberately conservative, and a survey wanting an unbiased estimator should not use it.

One objection remains: that Table 10 moves a single parameter and therefore overstates its importance. We repeated the sweep with m and s re-fitted on the 19 detector-training videos for *each* value of c , so the confidence condition is never blamed for a companion parameter left at a stale value. The conclusion strengthens. Given the freedom to compensate, the grid search responds to a lowered c by tightening the span constraint from 0 to 1 s, buying precision with duration instead, and the substitution does not pay. At $c = 0$ the held-out MAE is 3.69 against 2.38, and coverage falls to 60.2%, *below* the frozen variant, because a one-second minimum discards short tracks belonging to real animals. Across the full 378-cell grid only two values of c , 0.55 and 0.65, appear among the twelve best cells by fit MAE. The sweep also exposes a redundancy we report rather than hide. The three best cells are the same $c = 0.65$ rule with the span parameter at 0, 0.1 and 0.25, all giving identical fits. So s is *inert* at the operating point, and the published rule is effectively two parameters wearing a third.

Figure 8 lays out the full $m \times c$ plane with $s \geq 0$, error on the training videos beside error on the held-out ones. The held-out panel is for inspection only, since choosing a cell by it would be selecting on the test set. It answers the natural objection that a more permissive rule would simply remove the bottleneck. A rule of $m \geq 2$ and $c \geq 0.3$ never under-counts a held-out video, but it accepts 477 tracks where 83 animals exist, an MAE of 30.3, roughly six tracks per animal. The published cell is the single minimum of fit-set error. On held-out video the whole $c = 0.65$ column forms a flat valley, with MAE between 2.31 and 2.38 for every m from 2 to 20, so once c is set the minimum track length barely matters. The converse does not hold: a rule of length alone, $m \geq 40$ with no confidence condition, fits the training videos almost as well (MAE 2.4) but generalizes poorly (3.8). The confidence condition is what transfers. Changing how many frames the score averages leaves the picture unchanged. Re-fitting (m, c) with $k = 1, 2, 3, 5$ or 10 in Eq. (1) selects $c = 0.65$ every time, with held-out MAE between

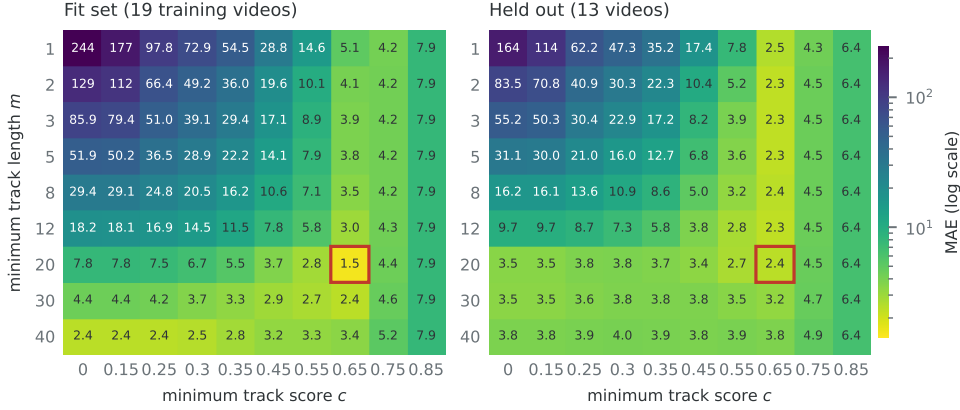


Fig. 8 Counting error across the confirmation-rule grid. MAE in animals per video, on a logarithmic colour scale shared by both panels; the published operating point ($m \geq 20$, $c \geq 0.65$) is outlined.

2.23 and 2.62, all inside the 95% interval of the published rule. Averaging over $k = 20$ frames reaches the lowest fit error and the worst held-out error, which is over-fitting. Fewer, higher frames fail outright: no held-out track’s two best frames average 0.85, so that rule accepts nothing. Every setting of the confirmation stage therefore trades missed animals against duplicates along a single curve. The published values sit at its minimum-error point on the training videos, and the learned confirmers of Section 4.7, which have no hand-set values at all, land on the same curve.

4.7 The Rule Is Not the Bottleneck

Section 4.6 shows that the rule’s parameters are load-bearing rather than arbitrary. The remaining question is whether the *form* of the rule limits the count. We tested a learned confirmer under three protocols and it does not. Because a negative result about learning is only as strong as the learners behind it, we state them in full. Every confirmer consumes the same input, one candidate track as a variable-length sequence of ten per-detection features. These are confidence, box centre, width, height, $\sqrt{\text{area}}$, the frame-to-frame deltas of centre and $\sqrt{\text{area}}$, and the time gap to the previous detection. Positions and sizes are normalised by frame size, so flicker appears as a large time gap and ego-motion as a systematic centre drift. Eight further features describe the track’s *competing group*, the tracks overlapping it in time whose mean boxes intersect it. They are the group size, the track’s length as a fraction of the longest in its group, and the group’s confidence and duration statistics. These are the features a duplicate must be separated on, since a duplicate differs from its primary not in appearance but in being the shorter, weaker member of a group. Appearance is excluded, because Section 4.11 shows it carries little signal at this pixel scale and because embeddings would let a 60k-parameter model memorise 32 videos.

The learners span three orders of magnitude in capacity. Logistic regression on pooled summary statistics has 13 parameters. Gradient boosting is run at depth 3

for 200 iterations (about 10^3 parameters) and as depth-1 stumps (about 40 effective parameters). The temporal transformer has 2 pre-norm encoder layers, $d_{\text{model}} = 64$, 4 heads, dropout 0.2 and sinusoidal position encoding. Masked attention pooling lets a 3-detection track and a 300-detection track pool comparably, and cross-track context is fused after pooling, for 6×10^4 parameters in all. Class imbalance, 207 primary tracks against 6801 duplicates and false candidates in pool C, is handled by inverse-frequency positive weighting in the loss ($w_+ = N_-/N_+ \approx 33$). Splits are by video throughout, reusing the detector’s assignment. We did *not* attempt hard-negative mining or cost-sensitive losses beyond that weighting, and we report this as a limitation. A better-tuned learner might close the gap, but the monotone capacity trend below is evidence about sample size rather than about tuning, and a tuning fix would have to explain why the *smallest* models do best.

Table 11 Held-out confirmation comparison. All methods trained and thresholded on the 19 detector-training videos, reported on the 13 held out. Learned confirmers lose on pool C and tie on pool G. Logistic regression on pool G fails in the opposite direction, predicting 102 animals where 83 exist.

Pool	Confirmer	MAE	Counted
C	Hand-tuned rule (3 params)	2.38	55/83
	Gradient boosting	2.85	—
	Logistic regression	3.00	—
	Temporal transformer ($\sim 60\text{k}$ params)	2.85	49/83
G	Hand-tuned rule	2.77	50/83
	Gradient boosting	2.92	—
	Logistic regression	3.15	—
	Temporal transformer	2.77	51/83

Table 11 gives the held-out comparison. Every learned method loses on pool C: gradient boosting and the temporal transformer at 2.85 MAE and logistic regression at 3.00, against the rule’s 2.38. On the richer pool G the transformer only ties, at 2.77. Every learned variant *under-counts harder* than the rule, so they reject additional candidates rather than recovering the animals the rule discards. Two further protocols agree. Under 8-fold cross-validation over videos the rule again wins, at 1.88 MAE; the per-fold search re-selects the published values in all eight folds, which is why this matches the all-video figure of Table 7. The pattern across that sweep is diagnostic. Performance declines monotonically with capacity: logistic regression at 13 parameters reaches 3.41, gradient boosting at 10^3 reaches 2.41–2.66, and the transformer at 6×10^4 reaches 2.81. Depth-1 stumps (about 40 effective parameters) recover nearly the whole gap and achieve the best root-mean-square error of any method. This is a variance limit, not an architectural one. We also trained directly on the counting objective, letting fragments share probability mass so the loss is $|\sum_i p_i - c|$ rather than per-track cross-entropy. It ranked last of the variants tested, which rules out “the rule wins because it optimizes MAE directly” as the explanation. The remaining sections

ask whether duplicate suppression can be improved directly, how the pipeline transfers across sites, and which animals are lost.

4.8 Duplicate Suppression at the Frame and Track Level

Since the negatives that limit counting are duplicates, the natural question is whether better duplicate suppression solves it. We tested the two levels at which suppression can act, and neither reaches the duplicates that matter. At the *frame* level, non-maximum suppression is already the mechanism, and Table 9 contains the sweep. Raising the NMS IoU threshold from 0.50 to 0.90, which suppresses less so that more overlapping boxes survive, multiplies the candidate pool thirteenfold and produces the worst count in the paper, 41 of 83. Frame-level suppression is therefore load-bearing in the direction of *more* suppression, not less. We did not test Soft-NMS [6] or Confluence [58], and the reason is structural rather than incidental. Both decide between boxes present in the same frame, while the duplicates that cost counts here are separate tracks on the same animal, often seconds apart, that no frame-level rule can see.

At the *track* level we did test clustering, which is the operation matching the problem: accept tracks that lie on an animal, then merge the accepted tracks into animals. Stage one becomes an easier learning problem, since “is this track on a deer at all” has 632 unambiguous positives instead of 207 contested ones. Stage two is geometry: tracks overlapping in time within a box-scale of each other are one animal, and the count is the number of clusters. It reaches MAE 3.66 against the rule’s 1.88 on the same 32-video protocol, and the error changes character rather than shrinking. Bias falls to -0.34 from -0.81 , but over-counts rise to 53 and under-counts to 64, against 17 and 43. The clustering is nearly unbiased and much noisier, which is what happens when merging decisions are made on geometry alone at 27 pixels. This is the same trade as Table 10: the methods that recover animals admit duplicates at a similar rate, and the published rule sits where the two costs balance.

4.9 Generalization Across Sites

The held-out protocol of Section 3.7 splits by video, so a held-out video may come from a site the detector has seen. Figure 9 asks the stricter question: train on three sites and count on the fourth, four times over, 8 videos per fold. Every one of the 235 detectable animals is then scored by a detector that never saw its site. The confirmation rule is *not* refitted per fold; the published (m^*, s^*, c^*) is applied unchanged, which is the honest reading of “does this pipeline transfer”.

Figure 9 puts the three stages on one set of axes, and the shapes carry the result. Detection and association hold a large and nearly regular quadrilateral across all four sites, while the counted polygon is lopsided and pinches to a point at MAS. Per-fold MAE is 7.38, 4.00, 3.38 and 1.88 animals per video for SHB, TON, SHW and MAS. Two things follow. First, detection transfers well but not perfectly. Pooled over the four folds, REACHED falls from the within-site 98.8% to $91.1 \pm 3.7\%$, PRIMARY from 88.0% to $78.3 \pm 5.2\%$, COUNTED from 66.3% to $51.9 \pm 6.4\%$, and MAE rises from 2.38 to 4.16 animals per video. These are measured over all 235 detectable animals rather than

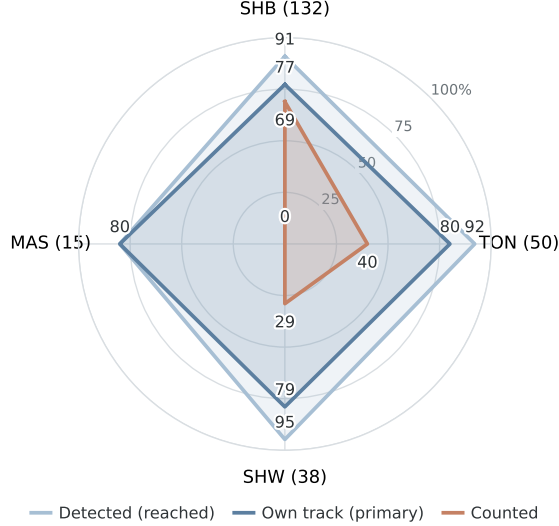


Fig. 9 Leave-one-site-out. Each axis is one held-out site, trained on the other three, as a percentage of that site’s animals; animal counts in parentheses.

the 83 of the main protocol, so they are the tighter estimates in the paper. The claim that detection is near-saturated should therefore be read as within-corpus; across sites a real detection gap opens, and Section 5 treats that as the boundary of the claim.

Second, COUNTED falls further and unevenly, from 69% of SHB’s animals down to none of MAS’s, and the per-fold score ceiling names the cause. The highest track score $\bar{c}_5$ any long track attains is 0.862 on the SHB fold, 0.782 on TON, 0.752 on SHW and 0.585 on MAS. The rule’s $c \geq 0.65$ is an *absolute* threshold on a detector score, and a detector trained without a site scores that site systematically lower. On the MAS fold no track exceeds 0.585, so a threshold at 0.65 accepts nothing, and the pipeline returns zero on 15 animals it detected 12 of. We cannot attribute that shift to site as such. Each site was surveyed on a single night, so a fold removes a site together with that night’s ambient temperature, humidity and wind, which set how far an animal’s heat signature stands above the background. The shift is therefore between deployment conditions, of which site is only one, and separating them would need the same site surveyed on several nights. This is the failure mode Section 4.6 already identified: the confidence condition is doing duplicate suppression on a score scale that itself depends on deployment conditions.

The obvious repair does not work, which is informative. A fold-relative threshold accepts the same *fraction* of long tracks that $c \geq 0.65$ accepts on pool C. It restores coverage to $80.9 \pm 5.1\%$ and destroys the count, at MAE 8.88 and a bias of +6.06 animals per video against -2.91 for the frozen rule. The cross-site folds generate more long tracks than the within-site pool, so a fixed acceptance fraction admits duplicates faster than it recovers animals. That is the inversion of Section 4.4 reappearing across

sites. Transferring this pipeline to a new site or survey night therefore requires re-fitting the threshold on labelled data from it, and quantifying how much labelled data that takes is the most practically valuable open question this corpus leaves.

4.10 Which Animals Are Missed

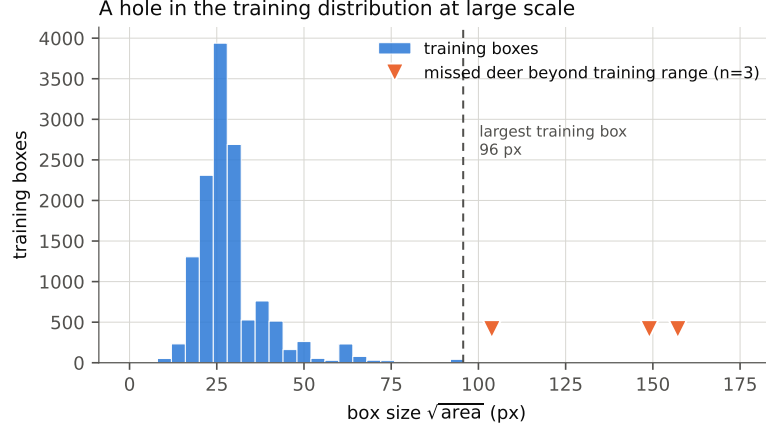


Fig. 10 A hole in the training distribution. The training split contains no box above 96 px. Three missed animals sit at 104, 149 and 157 px, one of them never detected at all. A detector that finds 27 px animals does not miss a 149 px one unless it has never seen one.

Of the 83 held-out animals, the detector never fires on 9 (median 20 px, at the sensor’s limit) and the rule rejects 27 that it did find. Of those 27, **26 fail the confidence condition** and one fails purely on track length. Their five best detections average below 0.65, though five do reach 0.65 in a single frame, so the failure is one of sustained rather than peak confidence. Those 26 are the entire remaining opportunity, and Section 4.6 shows they cannot be recovered by moving the threshold that rejects them, since doing so admits roughly two duplicates for each. We verified by direct inspection that none of the 36 are annotation errors: each was rendered with its ground-truth box at three points in its track, and all are real animals. The three groups are separated by degree rather than by kind, which is why Section 4.7 finds what it finds. Counted animals have a median box of 38.1 px, 102 annotated frames and candidate confidence 0.76; the rejected group sits at 28.0 px, 47 frames and 0.52; the undetected at 19.9 px and 47 frames. The rejected animals lie between the other two on every measure, so no threshold on any one of them separates them cleanly.

Turned around, the same data give the probability that an animal is found and counted as a function of its size, which is the form a survey needs. With each animal’s median box side as the range proxy of Section 3.8, the detector finds 78% of the 235 detectable animals below 20 px and essentially all of them above 30 px. Counting falls off far faster: 81% of animals at 30–40 px are counted, 57% at 20–30 px, where the corpus median sits, and 22% below 20 px. On the 83 held-out animals alone the

20–30 px figure is 36%. The loss that range causes is therefore mostly a confirmation loss rather than a detection loss. Above 100 px the counted rate falls back to 50%, the training-distribution gap described next. Box size also carries body size, pose and thermal contrast, so this is a size-based detection function rather than a strict distance function, but it is the empirical input a distance-sampling estimate of abundance would need.

That inspection produced an unexpected finding. Three of the missed animals are not small but very *large*: 104, 149 and 157 px against a corpus median of 27, and the 149 px animal was never detected at all. Figure 10 gives the likely explanation. The largest training box is 95.6 px and the split contains *zero* boxes at or above 100 px, so all three sit outside the annotated size distribution. Scale augmentation does present enlarged copies, but under the symmetric sampling used here an enlargement beyond $1.5\times$ is never drawn. The remedy is straightforward to state, and we do not evaluate it. Covering the upper tail requires sampling the resize factor from an explicit asymmetric range, so that a 96 px box can be presented at twice its size without the same draw shrinking a 27 px animal to sub-pixel. Ultralytics supports such a range from version 8.4. The loss is concentrated in any case: the one video containing them counts 3 of 8, three of its five misses being the oversized individuals, so this is worth roughly 4 points of the headline figure.

4.11 Appearance Carries No Identity Signal at 27 Pixels

Table 12 Identity matching is dominated by size, not appearance. All cues scored on the same 69 held-out animals. A two-number size descriptor reaches 0.638 and an ImageNet backbone scores *below* plain geometry. The trained encoder’s 0.899 is largely range matching, not identity. It is a probe for how much identity signal the imagery carries, not a component of the counting system: the appearance term in the pipeline uses the detector’s own backbone features.

Cue	Rank-1
Chance	0.101
Mean intensity alone	0.101
Thermal silhouette alone (scale removed)	0.435
Raw thermal crop	0.493
Box size alone (w, h)	0.638
ImageNet ResNet-50 features	0.725
Geometry (size, aspect, intensity)	0.797
Trained thermal encoder	0.899

A word on terms before the result. *Recognising an individual animal* is the task this subsection tests and rejects. *Appearance matching*, the ablated component that uses appearance to help associate detections within one video, is a different and much easier operation. The reported system performs neither (Section 3.3). Appearance-based recognition is the natural proposal for the association stage, and it is the one component we tested and *rejected on the evidence*. At this pixel scale appearance carries less identity information than a two-number bounding box, and a system built

on it would be reporting range, not identity. We evaluate closed-set identity matching, under the standard protocol, on **69 animals in 7 videos the detector never trained on**: each track is split temporally, the first half forms the gallery and the second half the query. Within video is the *ceiling*, since both halves share range, pose and thermal calibration, so a low score is decisive.

A trained thermal encoder reaches rank-1 0.899 ± 0.093 (Table 12), which sounds like working individual recognition until the decomposition. **Box size alone reaches** 0.638 ± 0.118 , and within a video an animal’s range changes slowly, so most of what looks like identification is matching *distance*. Appearance in isolation carries little signal: mean intensity carries exactly chance information, silhouette with scale removed reaches 0.435, and a pretrained CNN lands below plain geometry. At a median 29×24 pixels there is not enough detail to key an identity to, so we describe the system as performing *track association*, preventing double counting within a survey, and make no individual-identification claim. Appearance does help the *association* stage, which is a far easier problem. Enabling it gives the best candidate pool in the paper, 82 of 83 animals reached and 73 given their own track (Table 9), and that pool still counts three fewer animals than the baseline. Both halves of the conclusion are negative: appearance cannot identify individual deer at this resolution, and even where it improves association it does not improve the count. **The reported system uses neither appearance matching nor any form of individual recognition**, and we would advise against building a thermal survey pipeline on it until the imagery is several times larger than 27 pixels.

5 Discussion

5.1 Why Confirmation Resists Learning

The two negative results of Section 4.4 and Section 4.7 have a single explanation, and it is a property of the data rather than of the methods. The confirmation stage receives roughly two hundred positive tracks against 7000–28,000 negatives, and the negatives are not a homogeneous class. They divide into false candidates, which are easy, and *duplicate* tracks on animals already counted, which are not. A duplicate is a correct detection of a real animal. Section 4.10 shows that duplicates sit between the positives and the false candidates on every feature we measured: box size, track length and candidate confidence alike. A confirmer therefore receives contradictory supervision on exactly the examples that distinguish counting from detection. That is why capacity does not help. The capacity trend in Section 4.7 declines monotonically from 13 parameters to 6×10^4 , and shrinking a gradient-boosted model to depth-1 stumps recovers nearly the whole gap. It is also why every improvement to candidate generation costs count. Growing the pool grows the ambiguous middle of the negative class far faster than it grows the positives, so a rule tuned to survive the new ratio must be more aggressive, and aggression costs real animals. Extrapolating the capacity trend puts the crossover in the low thousands of positive tracks, an order of magnitude beyond this corpus.

5.2 Recall and Duplicate Suppression Are Bought with One Parameter

Section 4.6 makes the same mechanism visible in a single parameter. Deleting the confidence condition raises held-out coverage by 23 points, more than any other change we made. It also turns a systematic under-count into a systematic over-count, because roughly two duplicates are admitted for each genuine animal recovered. The condition is not filtering non-animals. It is the only cheap signal separating a first sighting from a re-sighting, and the recall it appears to discard is the price of that separation. This reframes what a practitioner should tune. The usual detection framing asks how low the confidence threshold can go before false positives appear, and under it $c = 0$ looks free, because coverage rises and the extra tracks are all on real animals. By Proposition 2 the question is instead at what point admitted duplicates equal missed animals, the point where the estimator’s bias crosses zero, which the next subsection locates and costs.

5.3 Choosing an Operating Point for Abundance Estimation

A survey ecologist and a per-transect report want different things from this system, and the confirmation threshold is where the difference is settled. Our headline rule minimises mean absolute error per video, and it under-counts: 58 animals predicted where 83 exist across the 13 held-out transects, a shortfall of 25. For an abundance estimate that is not the quantity to optimise. Setting $c = 0.45$ instead, where signed bias crosses zero in Table 10, makes the same pipeline predict **81 animals against 83**, an aggregate error of 2.4%. Individual transects are less accurate, with MAE rising from 2.38 to 3.38 animals per video, but the errors are now two-sided and cancel across a survey rather than accumulating in one direction.

Which is preferable depends on the downstream statistic, and the two are not interchangeable. A management unit’s population total is a sum over transects, so unbiasedness matters and per-transect variance largely averages out. A report of how many animals were seen on a particular road on a particular night is a per-transect quantity, and there the low-variance rule is correct. We report the minimum-error rule as our headline because it is the conventional choice for a counting system, and record the unbiased setting here because it is the one an abundance study should use. Our own headline metric obscures the distinction. The capped fraction of Eq. (7) deliberately prevents an over-count in one video from offsetting a miss in another, which suits a per-video counter but not a population total, where that cancellation is the point. Under the capped metric $c = 0.45$ scores 72.3%; under the aggregate it recovers 97.6% of the animals present.

One route we expected to work does not. The standard remedy for a known under-count is a sightability correction: estimate the detected fraction and divide by it. The held-out protocol makes that estimate impossible to obtain honestly from our own data. On the 19 detector-training videos the pipeline predicts 152 animals against 153, so a correction factor fitted there is 1.007, effectively no correction, while the held-out videos need 1.43. The contamination that inflates coverage on seen video also destroys any correction estimated from it. A sightability factor for this system would have to

be fitted on video the detector never trained on, which is to say on data that could have been used to train it instead.

A second limit on any abundance estimate lies upstream of the pipeline, in the ground truth. As in any visual survey, an annotated animal is one a reviewer could see in the footage, so every figure in this paper measures the pipeline against what is visible rather than against the animals present. Proposition 3 gives the null expectation for how many animals should be annotated at each box size if deer were spread evenly over the ground in view. We fitted its constant to the 91 animals annotated at 30–60 px. Within that range the law holds closely, the 30–40 and 40–60 px bins standing in a ratio of 1.39 against a predicted 1.40. Below about 25 px the annotation falls short of it: 29 animals are annotated at 15–20 px where Eq. (9) predicts about 210, and about 90 even under the strip law, the more conservative alternative. The annotated tracks do not shorten significantly with range (Kruskal–Wallis $p = 0.24$ across size quartiles), so a distant animal, once visible, is followed about as long as a near one. The likely cause is what the sensor can resolve. At 640×512 a distant deer is a few warm pixels, often partly behind vegetation or terrain, the same fall in detectability with range that spotlight counts face [64]. The comparison assumes deer are spread evenly with distance from the road, which a road transect cannot guarantee [16], and box size also varies with body size and posture (Eq. (8)). An abundance estimate built on this system would therefore need both factors: the pipeline’s size curve of Section 4.10 and a detection function for the survey itself.

5.4 Implications for Practice

What follows are consequences of the measurements above rather than recommendations drawn from survey experience, and they are scoped to what this corpus can support: one sensor, four sites, one winter, one species, 236 individuals. Within that scope, four findings run against the field’s default. Selecting a detector by its detection metric did not select the model that counts best here. The counting criterion re-ranks seven of eleven models by two or more places (Section 4.2), and the best model by AP_{50} counted worse than the fourth-best. Evaluating on video the detector had trained on inflated coverage by 15.5 points (Table 4). A larger candidate pool was not a better one in any of the four attempts we made (Table 9). And a learned confirmation stage did not beat a three-parameter rule at this data scale, with the capacity trend placing the crossover an order of magnitude beyond 236 individuals (Table 11).

Three observations point the other way. Contrast normalization was worth more than any architectural choice we tested, 0.519 against 0.299 test mAP_{50} , a larger effect than the entire 0.365–0.510 spread across eleven architectures. The upper tail of the object-scale distribution repaid the attention normally reserved for the lower one: three of our misses are animals *larger* than anything in the training split (Fig. 10), a failure mode a small-object-focused evaluation would not surface. And an absolute confidence threshold did not transfer between sites. A detector trained without a site scored that site systematically lower, and a threshold fitted elsewhere rejected every track on one fold (Section 4.9). This corpus cannot separate that shift from the different nights on which the sites were surveyed. Whether a small labelled sample from each new site or survey night is the right remedy is a question this corpus can pose but not answer.

5.5 Limitations

Several limits bound these claims. The ground truth contains only animals visible in the footage, and it thins out with range (Section 5.3), so every figure measures the pipeline against what the video shows rather than against the animals present. The corpus is small and narrow: 236 animals of one species from four sites in one winter, with one site contributing 56% of the animals, and a single tracker (BoT-SORT). Each site was surveyed on a single night, so site and night are confounded. Four of the 13 held-out videos are the detector’s validation split and influenced checkpoint selection (Section 3.7). The headline count is a per-video aggregate; the identity-matched figure is 10 points lower. No animal appears in more than one video, so cross-session identification could not be evaluated. Finally, the confidence threshold does not transfer across sites (Section 4.9), so deployment elsewhere requires re-fitting it on labelled local data.

6 Conclusions

TRACT counts white-tailed deer from thermal road transects, and decomposing its error moves the problem rather than solving it. What a practitioner would try to buy with a better detector is lost after detection, in the stage that decides whether a track is a new animal. Every improvement to detection and association that we tried left the count no better or worse, and learned confirmers of three capacities lost to a three-parameter rule. The reason is supervision rather than model class. The confirmation stage sees roughly two hundred positive tracks, and duplicates of animals already counted sit *between* the positives and the false candidates on every feature we measured. What separates a first sighting from a re-sighting is therefore precisely what a confirmer cannot learn at this scale. Two boundaries follow for anyone adopting the result. The pipeline does not transfer unaltered: a detector trained without one site and its survey night scores it low enough that an absolute confidence threshold can reject every track (Section 4.9). And the under-count is an operating-point choice rather than a limit, since at the threshold where signed bias crosses zero the same pipeline recovers 81 of the 83 held-out animals in aggregate (Section 5.3). We release the dataset, per-individual annotations, evaluation harness and decomposition tooling. The gap between what is reachable and what is counted is large, precisely localized and entirely open.

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Abbreviations.	AP	Average precision
	AUC	Area under the ROC curve
	CLAHE	Contrast-limited adaptive histogram equalization
	CVAT	Computer Vision Annotation Tool
	ECE	Expected calibration error
	FLIR	Forward-looking infrared
	GMC	Global motion compensation
	HOTA	Higher-order tracking accuracy
	IoU	Intersection over union
	LOSO	Leave-one-site-out
	MAE	Mean absolute error
	NMS	Non-maximum suppression
	RMSE	Root mean square error
	TRACT	Thermal Road-transect Animal Counting and Tracking
	UAV	Uncrewed aerial vehicle

Appendix A Complete Ablation Catalogue

Table A1 lists every ablation family run over the course of this work, including those measured at the detection stage rather than end to end. Families whose effect on the final count is quantified appear in Table 9; the remainder are reported in the sections cited beside them in the main text.

Table A1 Every ablation run. Thirteen families spanning preprocessing, detector choice, resolution, evaluation protocol, tracking, association and confirmation.

Ablation	Range Tested	Effect
CLAHE normalization	on / off	0.519 vs. 0.299 mAP ₅₀
Input resolution	640 / 1280	best detector counts worse
Detector architecture	12 models	0.365–0.510 AP ₅₀
Matching criterion	4 rules	re-ranks 7/11 models
Detector confidence	0.10 / 0.05 / 0.02	0 extra animals
Track-init threshold	0.15 / 0.05	+5 PRIMARY
Orphan recovery	on / off	+17 animals
NMS IoU	0.50 / 0.90	+4 PRIMARY, −14 COUNTED
Appearance matching	on / off	+7 PRIMARY, −3 COUNTED
Confirmer capacity	3–60k params	monotone decline
Confirmation threshold c	0.00–0.85, m, s free	c is load-bearing
Appearance cue	7 variants	size dominates
Track splitting	threshold sweep	+3 for 4k candidates

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